

Template Matching on TurtleBot - Deployment Guide

(Default Mode: Compressed Image Transport, Remote Camera)

Terminal Alpha - TurtleBot Core

```
ssh ubuntu@<TURTLEBOT_IP>
export ROS_DOMAIN_ID=92
source /opt/ros/humble/setup.bash
ros2 launch turtlebot3_bringup robot.launch.py
```

Terminal Bravo - TurtleBot Camera Node

```
ssh ubuntu@<TURTLEBOT_IP>
export ROS_DOMAIN_ID=92
source /opt/ros/humble/setup.bash
ros2 run camera_ros camera_node --ros-args -p format:=MJPEG -p width:=640 -p height:=480
```

Terminal Charlie - Laptop Template Matcher

```
export ROS_DOMAIN_ID=92
source /opt/ros/humble/setup.bash
source ~/ros2_ws/install/setup.bash
ros2 launch template_match template_match_test.launch.py
```

Terminal Delta - Laptop Call Matching Service

```
export ROS_DOMAIN_ID=92
source /opt/ros/humble/setup.bash
ros2 service call /perform_template_matching example_interfaces/srv/SetBool "{data: true}"
```

Terminal Echo - Optional RQT Viewer (Laptop)

```
ros2 run rqt_image_view rqt_image_view
-> Select /camera/image_raw/compressed
```

Alternate: Laptop Webcam Test Mode

```
ros2 launch template_match template_match_test.launch.py \
  use_local_camera:=true image_topic:=/camera/image_raw/compressed
```